CSE 4074 Computer Networks

Fall 2021 Midterm Exam

23.11.2021 Tuesday

Duration: 50 Minutes for Part 1, Total: 90 Minutes.

Q1	Q2	Q3	SUM
/63	/17	/20	/100

On my honor, I have neither given nor received any unauthorized assistance on this examination.

Name Surname:_____

Signature:_____

Solution and Submission Notes

- All notes are closed. You are not permitted to refer to any source during the exam.
- Write your name, surname and student ID on the top of each exam solution page.
- Before starting to the exam, fill all the required areas in the test answer sheet provided to you. Do not forget to mark your group (A or B).
- The exam consists of two parts.
- In the first part, there are 21 multiple choice questions. You have to finish this part in 50 minutes, and mark your answers on the test answer sheet. After 50 minutes, test answer sheets will be collected.
- Each correct answer will bring +3 points, while each wrong answer will cause -0.75 points.
- The second part consists of two open-ended questions. Show all your work.

Q-1. (63 pts) Multiple choice questions.

1) In DSL Access network, how voice (telephone) and data (internet) are separated from each other?

a. Voice is transmitted over phone line, while data is transmitted over a separate HFC cable.
b. Both voice and data goes to telephone net, then data is forwarded to Internet gateway.
c. Both voice and data goes to Internet, then voice is forwarded to PSTN network.
d. Voice and data are transmitted over DSL phone line at different frequencies.
e. Voice and data are transmitted over DSL phone line at different time slots.

2) What is an RFC?

a. It is a document that include all details of a standard protocol.
b. It is a standard protocol for remote field communication.
c. It is a standardization body for the Internet protocols.
d. It is a proprietary application layer protocol for video streaming.
e. It is a foundation responsible for allocating wireless spectrum.

3) Which of the following is/are true for packet switching

I. Packets may be dropped or queued.
II. Resources in the routers are reserved in advance.
III. Utilization is lower compared to circuit switching.
IV. Requires more complicated core network compared to circuit switching.

a. Only I
b. I, IV
c. I, II, IV
d. II, III, IV
e. I, III

4) Which of the following is true about Simple Mail Transfer Protocol (SMTP) and Hypertext Transfer Protocol (HTTP)

a. SMTP is a push-protocol, while HTTP is a pull-protocol.
b. HTTP can be used instead of SMTP.
c. SMTP can be used instead of HTTP.
d. SMTP is used for receiving e-mails from the mail-box, while HTTP is used for receiving web objects from a web server.
e. None of the above.

5) In a computer network, Host A is directly connected to host B. Transmission delay of sending a large file from A to B is 2 minutes. Which of the following would decrease the transmission delay?

I. Increasing the capacity of link between A and B.
II. Reducing the length of the link between A and B.
III. Compressing the file.
IV. Increasing the size of the receiver buffer in B.

a. Only I
b. Only II
c. I and III
d. I, II and III
e. I, II, III and IV

6) Suppose that you have a drone in your farm, which takes photos of the plants and send to you via LTE connection. LTE connection of the drone has 8 Mbps download capacity and 4 Mbps upload capacity. You are receiving photos to your laptop which has a WiFi connection with 10 Mbps download capacity and 5 Mbps upload capacity. What would be the maximum possible average end-to-end throughput while downloading the plant images?

a. 4 Mbps
b. 5 Mbps
c. 8 Mbps
d. 10 Mbps
e. 14 Mbps

7) In HTTP, the conditional GET mechanism

a. Imposes conditions on the objects to be requested.
b. Limits the number of response from a server.
c. Provides security by selectively sending requested objects according to client properties.
d. Helps to keep a cache up-to-date.
e. None of the above.

8) Suppose that Doctor Hüseyin Cerrah in Istanbul wants to do remote surgery for a patient in Parirenyatwa General hospital in Zimbabwe. The network delay should be less than 20 milliseconds for reliability purposes. Dr Cerrah realizes that the current delay is 100 ms. Which of the following will help Dr Cerrah to achive the desired delay of 20 ms. (Signal speed can be considered as 200.000km/sec. Access network capacity of the hospital is 100Mbps, while the access network capacity of Dr. Cerrah's own network is 200 Mbps).

a. 5x increase the access network capacity of the hospital.
b. 5x increase the access network capacity of Dr Cerrah's own network.
c. Use LEO satellite network.
d. Upgrade the communication technology to ultra reliable low latency 5G system.
e. None of the above. Dr. Cerrah should forget about remote surgery from Istanbul to Zimbabwe.

9) Which of the following is an example of IP spoofing?

a. Blocking all the packets originating from a specific IP address.
b. Changing destination IP address of an ongoing packet.
c. Intercepting DNS queries and returning bogus replies.
d. Bombarding DNS servers with too much traffic.
e. Reading contents of the packets that are sent from a source IP to a destination IP.

You own a company named HiTech Farming inc. Your domain name is hitechfarming, and the following table gives some DNS records related to your company. Some of the entries are not written. Questions 10, 11 and 12 are related to this table.

	Name	Value	Type	TTL (sec)
RR1	drone1.hitechfarming.com	196.133.35.123	A	86400
RR2	ns.hitechfarming.com	196.133.35.135	A	86400
RR3	mailserver.hitechfarming.com	196.133.35.132		86400
RR4	hitechfarming.com	mailserver.hitechfarming.com		86400
RR5	webserver.hitechfarming.com		A	86400
RR6	hitechfarming.com	ns.hitechfarming.com	NS	86400
RR7	www.hitechfarming.com	webserver.hitechfarming.com		86400

10) Which of the above resource records are stored in TLD DNS server?

a. RR1, RR3, RR4, RR5 and RR7

b. RR2 and RR6

c. Only RR2

d. RR4 and RR6

e. RR3 and RR5

11) What should be the types of RR3, RR4 and RR7, respectively

a. A, MX, NS

b. A, MX, CNAME

c. MX, CNAME, CNAME

d. MX, MX, CNAME

e. A, CNAME, CNAME

12) Which of the following is false?

a. If TTL for RR5 is decreased, customers will experience less interruption when IP address changes.

b. If TTL for RR5 is increased, this resource record will stay more in the Local DNS servers.

c. Value of RR5 cannot be 196.133.35.132 (same as RR3)

d. If you want to setup a new host with a new IP address, you need to add an A type resource record in authoritative DNS server, but you do not need to add anything in TLD server.

e. For hitechfarming.com, there could be multiple NS type resource records.

13) Which of the following is achieved by tit-for-tat mechanism in Bittorent P2P File sharing

a. Self-scalability.

b. Obtaining list of peers in the torrent.

c. Giving incentive to users to stay as an uploader even after they get the entire file.

d. Creating the torrent by gathering all users interested in the same file.

e. Dividing files into chunks.

14) Which of the following is false for Dynamic Adaptive Streaming over HTTP (DASH)?

a. Client determines when to request the chunk to avoid buffer starvation.

b. Client determines the quality of the chunks according to the available bandwidth.

c. Client determines which from which server to request chunks.

d. URLs of chunks are stored in a manifest file.

e. Available bandwidth is informed to the client by the server.

15) Which of the following is not an elastic application in terms of throughput?

a. File transfer

b. E-mail

c. Downloading web documents

d. text messaging

e. video streaming

16) DNS uses UDP instead of TCP. Which of the following is a reason for this?

I. Avoiding connection setup delay

II. Reducing the header size

III. Reducing the overhead in the DNS servers

a. Only I

b. I and III

c. I and II

d. II and III

e. I, II and III

17) What is the meaning of flow control?

a. Reducing the sending rate in case of network congestion.

b. Adjusting the sending rate according to the receiving rate.

c. Checking the network flows for security purposes.

d. Forwarding some of the packets in a network flow to the network controller.

e. Detecting errors in a data flow.

18) Which of the following is false for Content Distribution Networks (CDNs)

a. Multiple copies of the videos are stored at multiple geographically distributed sites.

b. The main aim is to bring services closer to the users with high availability.

c. Enter deep approach provides less delay compared to bring-home approach.

d. Enter deep approach enables simpler maintenance of the servers compared to bring-home approach.

e. CDNs often bypass tier-1 and regional Internet Service Providers.

19) Which of the following is true?

a. IP datagram is stored inside a TCP segment.

b. Port numbers and IP addresses are written in the IP header.

c. Checksum in the UDP header can be used to detect errors and to implement reliable data transfer.

d. A link layer frame is smaller than its corresponding IP datagram.

e. Link layer headers do not change through the path between the source and the destination.

20) Consider the following two 16-bit words: 1010101010101010, 1101010101010101. These words are received as X01010101010101Y and Z101010101010101. Which one of the following is a possible value of X, Y and Z, if the receiver does not detect any error using UDP checksum?

a. X=1, Y=1, Z=1

b. X=0 Y=1 Z=0

c. X=0 Y=0 Z=0

d. X=0 Y=1 Z=1

e. X=1 Y=1 Z=0

21) Which of the following is false?

a. HTTP is a stateless protocol.

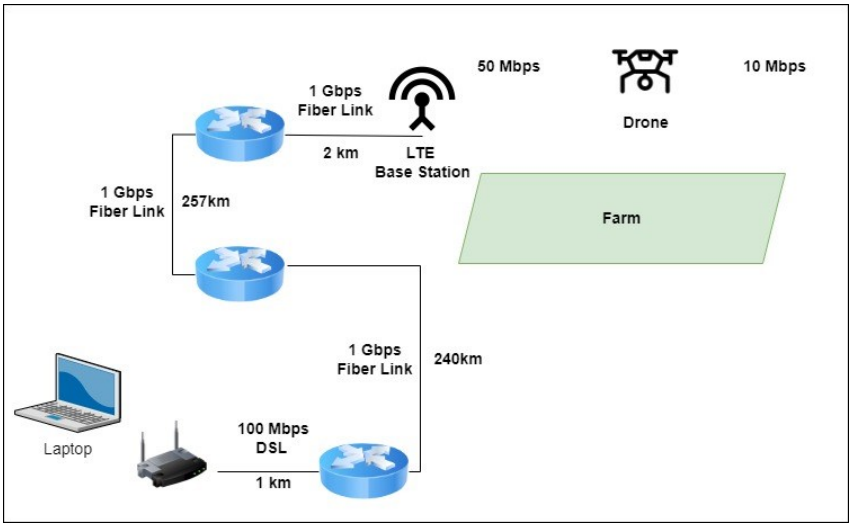
b. User states can be stored with the help of cookies.

c. Cookie ID is created by the browser and sent via cookie header.

d. Cookie file is managed by the user's browser.

e. User specific information is stored in backend database at the web site.

Q-2. (37 pts) Suppose that you have a farm in Manisa, and you want to use a drone to capture plant images and retrieve them via an application running in your remote laptop located in Istanbul. Also you want to control movements of the drone by sending messages from your laptop. Your laptop is connected to a WiFi access point, and the drone has an LTE connection. But depending on the distance to the LTE base station, access link capacity of the drone varies between 10 Mbps and 50 Mbps. Link lengths and capacities are given in the below figure. Network is idle, so queuing delay can be ignored. There is a total of 2 ms processing delay in the whole network (in both directions).



a) (10 pts) Suppose that you decided to use HTTP for sending commands (LEFT, RIGHT, FORWARD, BACKWARD, UP, DOWN) to the drone (also it is not the best way to communicate).
There is an HTTP server running on the drone and when it receives a "GET /drone1/LEFT HTTP/1.1" message, it turns left and sends back a "HTTP/1.1 200 OK" message.
It is required to get an answer from the drone in 10 ms. Is it possible achieve this via non-persistent HTTP connections? What about persistent HTTP connections? Describe your answer.

b) (6 pts) If there is a significant packet loss in the wireless links due to interference, what could be done to increase reliability of your system (do not forget that it is required to get an answer from the drone in 10 ms).

c) (8 pts) Lets say that the drone captures an image with a size of 125 Kilobytes and sends it to the laptop. What would be the maximum value of the total end-to-end delay to send the whole image from the drone to the laptop?

d) (5 pts) What could be done to reduce delay jitter (delay variation) while the drone captures images from different parts of the farm?

e) (8 pts) (This question is not related to the above parts). Suppose N packets arrive simultaneously to a link at which no packets are currently being transmitted or queued. Each packet is of length L and the link has transmission rate R. What is the average queuing delay for these N packets?